2.3 Abstract Algebra II: Problems

Attempt each problem for at least 30 minutes before reading solutions.md.

Warm-up

1. Show that $\mathbb{Z}[i]$ is a Euclidean domain with respect to $N(a+bi) = a^2+b^2$. Find its units, and factor 2, 3 and 5 into irreducibles in $\mathbb{Z}[i]$.

2. Show that the ideal $(2, x)$ in $\mathbb{Z}[x]$ is maximal but not principal. Conclude that $\mathbb{Z}[x]$ is not a PID.

3. Show that the following are irreducible over $\mathbb{Q}$:

(a) x^4+1; (b) x^3-3x-1; (c) x^4+x+1.

4. Show that $[\mathbb{Q}(\sqrt{2},\sqrt{3}) : \mathbb{Q}] = 4$, that $\mathbb{Q}(\sqrt{2},\sqrt{3}) = \mathbb{Q}(\sqrt{2}+\sqrt{3})$, and find the minimal polynomial of $\sqrt{2}+\sqrt{3}$ over $\mathbb{Q}$.

5. List all abelian groups of order 72 up to isomorphism, in both invariant factor and elementary divisor form.

6. Construct a field with 4 elements as a quotient of $\mathbb{F}_2[x]$. Write out its multiplication table and show its multiplicative group is cyclic.

Core

7. Let R be a PID. Show that every non-zero prime ideal of R is maximal. Deduce that $F[x,y]$ (F a field) is not a PID.

8. Let $G = \mathbb{Z}^3/N$, where N is the subgroup generated by the columns of

$$A = \begin{pmatrix} 2 & 4 & 4 \\ -6 & 6 & 12 \\ 10 & -4 & -16 \end{pmatrix}.$$

Compute the Smith normal form of A and identify G in invariant factor form.

9. (a) A 6×6 complex matrix A has characteristic polynomial $(x-2)^4(x-3)^2$ and minimal polynomial $(x-2)^2(x-3)$. List all possible Jordan canonical forms, and give the invariant factors (rational canonical form) in each case.

(b) Show that two 3×3 matrices over a field F are similar iff they have the same characteristic and minimal polynomials. Show this fails for 4×4 matrices.

10. Let $A \in M_n(\mathbb{C})$ satisfy $A^m = I$ for some $m \geq 1$. Show that A is diagonalisable. Show by example that this fails over $\mathbb{F}_p$.

11. Let $\alpha \in \mathbb{R}$ be a root of $f(x) = x^3-3x-1$.

(a) Show that $2-\alpha^2$ is also a root of f, and deduce that $\mathbb{Q}(\alpha)$ is the splitting field of f over $\mathbb{Q}$.
(b) Find $\mathrm{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q})$, and confirm your answer using the discriminant.

12. Let F be any field and G a finite subgroup of $F^\times$. Show that G is cyclic.

13. Show that $\mathbb{F}_{p^n}$ contains a subfield with p^m elements iff $m \mid n$, and that such a subfield is unique.

14. (a) Show that $x^{p^n} - x$ is the product of all monic irreducible polynomials in $\mathbb{F}_p[x]$ whose degree divides n.

(b) Find the number of monic irreducible polynomials of degree 4 over $\mathbb{F}_2$, and list them.

15. (a) Show that every irreducible polynomial over a field of characteristic 0 is separable.

(b) Let $F = \mathbb{F}_p(t)$, the field of rational functions over $\mathbb{F}_p$. Show that $x^p - t \in F[x]$ is irreducible but not separable.

16. Let $\zeta = e^{2\pi i/5}$.

(a) Show that $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$ and $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ is cyclic of order 4.

(b) Find the unique subfield E with $[E : \mathbb{Q}] = 2$, in the form $\mathbb{Q}(\sqrt{d})$.

17. Show that the splitting field K of $x^4 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[4]{2}, i)$, that $[K : \mathbb{Q}] = 8$, and that $\mathrm{Gal}(K/\mathbb{Q})$ is the dihedral group of order 8.

18. Show that $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois with cyclic Galois group of order n generated by the Frobenius map $\varphi(a) = a^p$.

19. Show that the fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ are not isomorphic.

Challenge

20. Show that $f(x) = x^5 - 6x + 3$ has Galois group S_5 over $\mathbb{Q}$, and conclude that the roots of f cannot be expressed by radicals. (Prove the group-theoretic lemma you need: a p-cycle and a transposition generate S_p for p prime.)

21. Using the fact that every constructible real number has degree a power of 2 over $\mathbb{Q}$, show that:

(a) the angle 60° cannot be trisected with straightedge and compass;

(b) the regular 9-gon cannot be constructed with straightedge and compass.

22. (Kummer extensions.) Let F be a field of characteristic 0 containing a primitive n-th root of unity ζ, let $a \in F$, and let $K = F(\beta)$ where $\beta^n = a$. Show that K/F is Galois and that $\mathrm{Gal}(K/F)$ is cyclic of order dividing n.

23. (Primitive element theorem.) Let F be an infinite field and $K = F(\alpha, \beta)$ a finite separable extension. Show that $K = F(\alpha + c\beta)$ for all but finitely many $c \in F$. Deduce that every finite separable extension of an infinite field is simple.

24. Let R be a PID.

(a) Show that a finitely generated torsion-free R-module is free.

(b) Show that $\mathbb{Q}$ is a torsion-free $\mathbb{Z}$-module that is not free. What does this say about whether $\mathbb{Q}$ is finitely generated?